Deep Learning Viva Q&A

CO1: Understand the basics of Deep Learning and apply the tools to implement deep learningapplications
CO2: Evaluate the performance of deep learning models (e.g., with respect to the bias-variance trade␂off, overfitting and underfitting, estimation of test error).
CO3: To apply the technique of Convolution (CNN) and Recurrent Neural Network (RNN) forimplementing Deep Learning models
CO4: To implement and apply deep generative models.
CO5: Construct and apply on-policy reinforcement learning algorithms
CO6:To Understand Reinforcement Learning Process unit-1 Foundations of Deep learning unit2- Deep Neural Networks(DNNs) unit3-Convolution Neural Network(CNN) unit4- Recurrent Neural Network(CNN) unit5-Deep Generative Models unit6-Reinforcement Learning bro as per previous give me first que

Unit 1: Foundations of Deep Learning — Most Important Viva Questions with Answers

1. What is Deep Learning?

Deep Learning is a subset of Artificial Intelligence and Machine Learning that uses neural networks with multiple hidden layers to learn patterns from data automatically. It is mainly used in image recognition, speech processing, NLP, and recommendation systems.

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2. Difference between AI, ML, and Deep Learning?

Artificial Intelligence is the broader concept of making machines intelligent. Machine Learning is a subset of AI where systems learn from data. Deep Learning is a subset of ML that uses deep neural networks for complex learning tasks.

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3. What is an Artificial Neural Network (ANN)?

ANN is a computing model inspired by the human brain. It consists of neurons connected through layers such as input layer, hidden layer, and output layer to process and learn data patterns.

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4. What are the layers in Neural Network?

A neural network mainly contains:

- Input Layer → receives data
- Hidden Layer → performs computation and feature extraction

- Output Layer → gives final prediction or result

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5. What is an Activation Function?

An activation function decides whether a neuron should activate or not. It introduces non-linearity into the neural network so that complex problems can be solved.

Examples:

- Sigmoid
- ReLU
- Tanh

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6. What is ReLU?

ReLU stands for Rectified Linear Unit. It outputs the input directly if it is positive; otherwise, it outputs zero. It is widely used because it trains models faster.

$f(x) = \max(0, x)$

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7. What is Sigmoid Function?

Sigmoid is an activation function that converts values between 0 and 1. It is commonly used in binary classification problems.

$\sigma(x) = \frac{1}{1+e^{-x}}$

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8. What is Forward Propagation?

Forward propagation is the process where input data passes through the network layer by layer to generate output predictions.

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9. What is Backpropagation?

Backpropagation is the process of updating weights in the neural network by calculating errors and minimizing loss using gradient descent.

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10. What is Gradient Descent?

Gradient Descent is an optimization algorithm used to minimize the loss function by updating weights in the direction of minimum error.

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11. What is Epoch in Deep Learning?

One epoch means one complete pass of the entire training dataset through the neural network during training.

12. What is Batch Size?

Batch size is the number of training samples processed before updating model weights.

13. What is Loss Function?

Loss function measures the difference between actual output and predicted output. Lower loss means better model performance.

14. What is Overfitting?

Overfitting occurs when the model learns training data too well and performs poorly on new unseen data.

15. What is Underfitting?

Underfitting occurs when the model cannot learn the training data properly, resulting in poor accuracy.

16. Difference between Overfitting and Underfitting?

- Overfitting → High training accuracy but low testing accuracy
- Underfitting → Low accuracy on both training and testing data

17. What is Bias and Variance?

Bias is the error due to wrong assumptions in the model. Variance is the error due to excessive sensitivity to training data.

18. What is Deep Neural Network (DNN)?

DNN is a neural network having multiple hidden layers between input and output layers for learning complex patterns.

19. Advantages of Deep Learning?

- Automatic feature extraction
- High accuracy
- Handles large datasets
- Useful in image, speech, and text processing

20. Applications of Deep Learning?

Applications include:

- Image Recognition
- Speech Recognition
- Self-driving Cars
- Medical Diagnosis
- Chatbots and NLP systems

CO1: Understand various Parallel Paradigm CO2: Design and Develop an efficient parallel algorithm to solve given problem CO3: Illustrate data communication operations on various parallel architecture CO4: Analyze and measure performance of modern parallel computing systems CO5: Apply CUDA architecture for parallel programming CO6: Analyze the performance of HPC applications unit1-Introduction to Parallel Computing unit2-Parallel Algorithm Design unit-3 Parallel Communication unit4- Analytical Modeling of Parallel Programs unit5-CUDA Architecture unit6-High Performance Computing Applications

Unit 1: Introduction to Parallel Computing — Important Viva Questions with Answers

1. What is Parallel Computing?

Parallel computing is a technique where multiple processors execute different parts of a program simultaneously to solve problems faster and improve performance.

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2. What is the need for Parallel Computing?

Parallel computing is needed to solve large and complex problems quickly, reduce execution time, and improve system performance in scientific and real-time applications.

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3. What is a Parallel Program?

A parallel program divides a task into smaller subtasks that run simultaneously on multiple processors.

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4. Difference between Sequential and Parallel Computing?

- Sequential Computing executes one instruction at a time.
- Parallel Computing executes multiple instructions simultaneously using multiple processors.

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5. What are the types of Parallelism?

Main types are:

- Data Parallelism
- Task Parallelism
- Pipeline Parallelism

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6. What is Data Parallelism?

In data parallelism, the same operation is performed on different parts of data simultaneously using multiple processors.

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7. What is Task Parallelism?

In task parallelism, different tasks or operations are executed simultaneously on different processors.

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8. What is Flynn's Classification?

Flynn's Classification categorizes computer architectures based on instruction and data streams.

Types:

- SISD
- SIMD
- MISD
- MIMD

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9. Explain SISD Architecture.

SISD stands for Single Instruction Single Data. It is a traditional sequential computer where one processor executes one instruction on one data stream at a time.

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10. Explain SIMD Architecture.

SIMD stands for Single Instruction Multiple Data. One instruction operates on multiple data elements simultaneously. It is commonly used in image processing and GPUs.

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11. Explain MIMD Architecture.

MIMD stands for Multiple Instruction Multiple Data. Multiple processors execute different instructions on different data simultaneously.

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12. What is Speedup in Parallel Computing?

Speedup measures how much faster a parallel program performs compared to a sequential program.

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$$Speedup = \frac{Sequential\ Time}{Parallel\ Time}$$

13. What is Efficiency?

Efficiency measures the effective utilization of processors in a parallel system.

$$Efficiency = \frac{Speedup}{Number\ of\ Processors}$$

14. What is Scalability?

Scalability is the ability of a parallel system to maintain performance when processors or workload increase.

15. What is Amdahl's Law?

Amdahl's Law states that the speedup of a parallel program is limited by the sequential portion of the program.

$$Speedup = \frac{1}{(1-P)+\frac{P}{N}}$$

16. What is Granularity in Parallel Computing?

Granularity refers to the amount of computation performed between communication operations.

- Fine Grain → More communication
- Coarse Grain → Less communication

17. What is Load Balancing?

Load balancing distributes work equally among processors to avoid processor idle time and improve efficiency.

18. What is Synchronization?

Synchronization coordinates multiple processes or threads to ensure correct execution order and avoid conflicts.

19. What is Distributed Memory Architecture?

In distributed memory systems, each processor has its own private memory and processors communicate using message passing.

20. What is Shared Memory Architecture?

In shared memory systems, multiple processors share the same memory space for communication and data access.

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21. Applications of Parallel Computing?

Applications include:

- Weather Forecasting
- Scientific Simulations
- Artificial Intelligence
- Image Processing
- Big Data Analytics
- High Performance Computing (HPC)